Project Journal - Milestone 1: Abusive Language Detection- Jaime Morales.

Topic: Abusive Language Detection in Social Media

1. Introduction and Problem Definition

Abusive language on social media has evolved from simple insults to complex forms of cyberbullying, hate speech, and harassment. The challenge is no longer just identifying "bad words," but understanding the context, intent, and cultural issues.

As an individual project, my focus is on **Domain Transfer**. Most models perform exceptionally well on the dataset they were trained on (e.g., Twitter) but fail when applied to different platforms (e.g., Reddit or news comments) due to changes in slang, length of text, and community norms.

2. State-of-the-Art (Literature Review)

To establish a solid foundation, I have analyzed two key academic pillars:

- **Reference 1: Rosa, H., et al. (2019). "Automatic cyberbullying detection: A systematic review."** This paper provides a comprehensive taxonomy of abusive behavior. It highlights that the most effective features have transitioned from manual "BoW" (Bag of Words) to deep learning. However, it identifies a major gap: the lack of generalization across different social media environments.
- **Reference 2: Waseem, Z., et al. (2017). "Understanding Abuse: A Typology of Abusive Language Detection."** This work is crucial for the project as it discusses the sub-tasks of abuse (hatespeech vs. personal attacks). It suggests that models need to move beyond surface-level patterns to understand the "target" of the abuse.

3. Identification of Best Existing Datasets

For this project, I will utilize and compare the following datasets to test domain transfer:

1. **Hatebase (Twitter):** A large-scale dataset focused on hate speech with specific labels for different types of discrimination.
2. **Jigsaw Toxicity Dataset (Civil Comments):** Unlike Twitter's short-form text, this contains longer, more structured comments from news sites.
3. **OLID (Offensive Language Identification Dataset):** Used in SemEval, it provides a hierarchical labeling system (offensive vs. non-offensive, targeted vs. untargeted).

4. Proposed Solution and Technologies

I propose a solution based on **Transfer Learning** using a **RoBERTa (Robustly Optimized BERT approach)** architecture.

- **Algorithm:** I will use a pre-trained Transformer model and fine-tune it on a source domain (e.g., Jigsaw).
- **Tools:** Python, Hugging Face Transformers library, and PyTorch.

5. Personal Opinion and Potential Improvements

Most current solutions are "black boxes" that rely too much on specific toxic keywords. In my opinion, the next step is **Contextual Awareness**.

I want to see if a model can learn the concept of abuse independently of the platform's specific vocabulary. Another improvement is the mitigation of **False Positives** in reclaimed language (where minority groups use "slang" terms that a model might wrongly flag as abusive).

6. Project Timeline & Next Steps

- Data preprocessing and cleaning (handling emojis, hashtags, and social media noise).
- Baseline model training and domain transfer tests.